CLAUDE HANDOFF GUIDE: HOW TO WRITE CHAPTERS 4, 5 AND 6 FROM THIS PACKAGE

> **Audience.** An AI assistant or the author, writing the three chapters from this evidence set. Repository `student-beneficiary-system`, branch `fix/sbms-completion-work`, commit `d6b5032`. Nothing in this package refers to any other commit unless it says so.

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1. Read in this order

Order	File	Why
1	`ACADEMIC_GUIDE_REQUIREMENTS.md`	The structure to write into, and the six ambiguities you must not resolve by invention
2	`SBMS_EVIDENCE_PROVENANCE_AND_CLAIM_LIMITS.md`	What may not be claimed. Read this before drafting a single sentence
3	`CHAPTER_4_…_EVIDENCE.md`, `CHAPTER_5_…_EVIDENCE.md`, `CHAPTER_6_…_EVIDENCE.md`	The evidence for each chapter, with the section pointers
4	The drafted chapter you are editing (`CHAPTER_4_…`, `CHAPTER_5_…`, `CHAPTER_6_…`)	The prose already exists; improve it rather than rewriting from scratch
5	`MASTER_EVIDENCE_REGISTER.md`	Evidence IDs and statuses for any claim you add
6	`CROSS_CHAPTER_CONSISTENCY_AND_TRACEABILITY.md`	Keeps Chapters 4–6 consistent with 1–3 and with each other
7	`DIAGRAM_SPECIFICATIONS.md`, `FIGURE_PLACEHOLDER_CHECKLIST.md`	Figures: what must be drawn or captured
8	`ACADEMIC_COMPLETENESS_AND_GAP_AUDIT.md`, `DOCUMENTATION_GAP_REGISTER.md`	What is still missing, ranked by risk

The twelve deep catalogues (the `SBMS_…` files with suffixes `CATALOGUE`, `DOCUMENTATION`, `GUIDE` and `INVENTORY`) are reference material: consult them per section, do not summarise them wholesale into the chapters.

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2. The file set and what each file is for

File	Role
`ACADEMIC_GUIDE_REQUIREMENTS.md`	Guide structure, compliance matrix, ambiguities
`CHAPTER_4_SYSTEM_ANALYSIS_AND_DESIGN_EVIDENCE.md`	Chapter 4 evidence entry point
`CHAPTER_5_IMPLEMENTATION_TESTING_AND_EVALUATION_EVIDENCE.md`	Chapter 5 evidence entry point
`CHAPTER_6_CONCLUSIONS_RECOMMENDATIONS_AND_FUTURE_WORK_EVIDENCE.md`	Chapter 6 evidence entry point
`CROSS_CHAPTER_CONSISTENCY_AND_TRACEABILITY.md`	Consistency matrix, traceability chains, contradictions
`DIAGRAM_SPECIFICATIONS.md`	All 24 figures, attributes and drawing rules
`MASTER_EVIDENCE_REGISTER.md`	51 evidence IDs with locations and statuses
`ACADEMIC_COMPLETENESS_AND_GAP_AUDIT.md`	Coverage, missing evidence, secret check
`CLAUDE_HANDOFF_GUIDE.md`	This file
`CHAPTER_4_SYSTEM_ANALYSIS_AND_DESIGN.md`	The drafted chapter (1 176 lines)
`CHAPTER_5_IMPLEMENTATION_TESTING_AND_EVALUATION.md`	The drafted chapter (838 lines)
`CHAPTER_6_CONCLUSIONS_RECOMMENDATIONS_AND_FUTURE_WORKS.md`	The drafted chapter (322 lines)
`SBMS_COMPREHENSIVE_CHAPTERS_4_5_6_INFORMATION_REPORT.md`	The consolidated evidence report, §1–§29
`SBMS_REQUIREMENTS_AND_SCOPE_CATALOGUE.md`	37 FRs, 21 NFRs, business rules
`SBMS_SYSTEM_ARCHITECTURE_AND_MODULE_MAP.md`	Modules, flows, decisions

File	Role
`SBMS_DATABASE_SCHEMA_AND_DATA_DICTIONARY.md`	13 tables, columns, state machines
`SBMS_API_ENDPOINT_CATALOGUE.md`	45 endpoints
`SBMS_ML_SUBSYSTEM_DOCUMENTATION.md`	Features, artifact, metrics, limits
`SBMS_TEST_EVIDENCE_CATALOGUE.md`	Every test artefact with date and command
`SBMS_SECURITY_AND_ACCESS_CONTROL_CATALOGUE.md`	Controls and residual risk
`SBMS_DEPLOYMENT_AND_OPERATIONS_GUIDE.md`	Azure resources, pipeline, runbook
`SBMS_UI_SCREEN_AND_WORKFLOW_INVENTORY.md`	Screens and workflows
`SBMS_EVIDENCE_PROVENANCE_AND_CLAIM_LIMITS.md`	Evidence tiers and forbidden claims
`OBJECTIVE_REQUIREMENT_TEST_TRACEABILITY.md`	Objective → requirement → test → conclusion
`DOCUMENTATION_GAP_REGISTER.md`	Item-level gaps G-01…G-28
`EVIDENCE_AND_VERIFICATION_REPORT.md`	Claim-level inspection record
`FIGURE_PLACEHOLDER_CHECKLIST.md`	Capture and insertion checklist

Two tiers, deliberately: the first nine files are the entry set a writer reads; the rest is the evidence they point into. Do not duplicate content between tiers — cite it.

3. Non-negotiable writing rules

1. **No claim without a location.** Every factual sentence must trace to a file, route, migration, test or log, or to an evidence ID. If it cannot, it is deleted, not softened.
2. **Use the six statuses exactly.** VERIFIED AND IMPLEMENTED, VERIFIED BUT NOT IMPLEMENTED, TESTED AND PASSED, TESTED AND FAILED, NOT TESTED, UNKNOWN / REQUIRES CONFIRMATION. No seventh status, no "mostly".
3. **Counts carry dates and commands.** "151 tests" is wrong; "151 tests, `npx jest --runInBand tests`, 2026-11-01" is right. Never add counts across levels.
4. **Synthetic means synthetic.** Every ML metric is pair-level on generated data. Never write "the model is accurate", "the system detects duplicates correctly in practice", or any real-world accuracy claim.
5. **The human gate is the control.** Never describe the model as deciding a duplicate.
6. **Gaps stay visible.** FR-34…FR-37 and the partial requirements are reported, never absorbed into a success narrative or moved to future work to look complete.
7. **Deviations are stated once and kept.** The Random Forest → XGBoost change (G-21) appears in Chapters 4, 5 and 6 in the same words.
8. **Do not renumber figures or tables.** Figures 4.1–4.24, tables as numbered in the drafts.
9. **Do not invent citations.** Chapters 4–6 add no literature claims; a needed citation belongs in Chapters 1–3 and must be verified by the author.
10. **Do not print secrets.** No key, token, password, connection string or real personal data, ever.

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4. Drafting workflow per chapter

Step	Action
1	Open the chapter draft and the matching `…_EVIDENCE.md` side by side
2	For each section, copy the evidence row, not the prose: the fact, the location, the limitation
3	Write the sentence so that the limitation survives editing. If the limitation would have to be cut for the sentence to read well, the sentence is too strong
4	Check the figure callouts against `DIAGRAM_SPECIFICATIONS.md`; every figure number used must exist there
5	Check requirement numbers against `SBMS_REQUIREMENTS_AND_SCOPE_CATALOGUE.md` §6–§8; never invent an FR or NFR number

Step	Action
6	Re-read `ACADEMIC_COMPLETENESS_AND_GAP_AUDIT.md` §4 to confirm you have not silently closed a gap
7	Run the five-minute self-check in §6 below

5. Phrases to use, and phrases to avoid

Avoid	Write instead
"The model accurately detects duplicates"	"The model ranks candidate pairs; an administrator decides"
"95% accuracy"	"Precision 0.9204, recall 0.8927 on a synthetic test partition at a 0.50 cut-off"
"The system was validated with users"	"No user acceptance study was conducted (G-06)"
"The system is deployed in production"	"A demonstration instance is deployed on Azure App Service"
"All tests pass"	"24 suites / 151 tests passed on 2026-11-01 under the command shown"
"The QR code was tested on a phone"	"QR verification was exercised in a headless browser; no phone capture exists (G-11)"
"Random Forest-based duplicate detection"	"An XGBoost classifier over six features; the proposal's Random Forest survives as a research candidate"
"Requirements fully met"	"24 of 30 delivered requirements verified, 6 partial, 4 in-scope gaps"

6. Five-minute self-check before handing a chapter back

1. Every figure and table referenced exists and is numbered consistently.
2. Every number in the chapter appears in `MASTER_EVIDENCE_REGISTER.md`, `SBMS_TEST_EVIDENCE_CATALOGUE.md` or `SBMS_ML_SUBSYSTEM_DOCUMENTATION.md` with the same value.
3. No sentence claims real-world ML accuracy, user acceptance, production assurance or phone-based scanning.
4. Every gap named in the chapter matches its ID in `DOCUMENTATION_GAP_REGISTER.md`.
5. The chapter's conclusions are no stronger than Chapter 5's evidence.

7. When evidence is missing

Do not infer, and do not write around the hole. Do one of three things, in this order:

1. **Cite the gap** if it already has an ID (`DOCUMENTATION_GAP_REGISTER.md`).
2. **Add a gap row** if it is new: what is missing, why it matters, which section depends on it, who can close it.
3. **State the limitation in the sentence itself** if the section cannot be written without touching the gap — for example "the workstation specification was not recorded (G-01)".

An openly stated gap is academically acceptable. An invented number is not, and it is the one error that cannot be repaired after submission.

8. What this package does not do

- It does not draw the figures, capture the screenshots or run a user study.
- It does not edit application code, schema, migrations, configuration, model artifacts or deployment infrastructure.
- It does not resolve contradictions in favour of the more convenient version; it records them.
- It does not replace the author's judgement on the six ambiguities in `ACADEMIC_GUIDE_REQUIREMENTS.md` §6.

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9. PDF rendering

`docs/final-documentation-content/pdf/` holds a PDF of every file in this folder, plus `SBMS_CHAPTERS_4_5_6_DOCUMENTATION_PACKAGE.pdf`, which carries a cover page, a contents list and all 27 files in the §1 reading order. The PDFs add no content: they are this markdown, paginated, so a claim is no stronger in PDF than it is in the `.md` source.

Regenerate after any edit:

```
node scripts/docs-to-pdf.mjs            # every file, then the combined package
node scripts/docs-to-pdf.mjs --combined # combined package only
```

The renderer uses headless Chrome and no third-party packages, so a PDF can be no more than one command out of date with its source.